NUI Galway
OÉ Gaillimh

Semester 1 Examinations 2010/2011

Exam Code(s)	1MF1, 1SD1, 2CS1
Exam(s)	Higher Diploma in Applied Science (Software Design & Development), M.Sc in Software Design & Development, BSc (Computing Studies)
Module Code(s)	CT853, CT232
Module(s)	Algorithmics & Logical Methods, Methodology
Paper No.	1
External Examiner(s)	Prof. Michael O'Boyle
Internal Examiner(s)	Dr. Jim Duggan Martina Fox

Instructions: Answer any 3 questions. All questions carry equal marks

Duration	2 hours
No. of Pages	4
Discipline(s)	Information Technology
Course Co-ordinator(s)	

Requirements:
MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Release to Library: Yes ☐ No ☐

Q1 a Using the following methods write down(step by step) the position of each letter using the word ‘Seoul’ when sorted alphabetically using :

1. Selection Sort
2. Merge sort [10]

b What does the complexity of an algorithm mean? [5]

c. Of the following sorts -Bubble, Selection, Insertion, Merge, Quick-which has the best:

1. Best case Complexity?
2. Average case Complexity? [10]

Q2 a What is an algorithm and why do we study them? [5]

b What are the desirable properties of an algorithm [5]

c Develop a simple algorithm to find the location of the maximum of the positive integers stored in the array b[0..N-1], N > 0. You may assume that the required maximum is unique. As an example, in the array b[0..3], which contains the positive integers 4, 6, 3, 10 in that order, your algorithm should report that the maximum is to be found in position 3. [15]

Q3 a Explain the Divide & Conquer approach to algorithm design. Give an example (using pseudocode) of a Divide & Conquer algorithm to calculate X^n , where n is a positive integer [10]

b In cryptography we often need to multiply 2 large integers. Using a Divide and Conquer strategy, the Karatsuba algorithm gives us an efficient way to do this. Describe the algorithm explain why it is more efficient than ‘standard’ multiplication and give (in ‘big oh’ notation) its complexity

[15]

Q4 A truck has a total (weight) capacity of 4 Kilo Tonnes. In port, there are four loads it can collect, each with a different weight and value:

Load	**A**	**B**	**C**	**D**
Weight (Kilo Tonnes)	2	1	4	3
Value (Millions of Euro)	5	3	7	6

The objective is to decide which loads to take to maximise the *Value* , while not surpassing the 4 Kilo Tonne *Weight* limit

a Using a Greedy approach to this problem, what is the solution obtained if the Greedy approach is based on:

1. Maximising value?
2. Minimising weight?
3. Maximising the (value/weight) ratio? [15]

b Describe (by composing a table of values) how Dynamic Programming can be used to solve this problem and state the solution obtained.

[10]